STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-GRADED-ALGEBRAS-TI

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 880-890 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 892-984, the complete author proof (91 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 2 of queue_examples-defos.json (pb=3188, stmt 880-890, proof 892-984).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0D3L: sources/stacks-examples-defos/pinned-original-source.tex, source lines 825–835. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Example 0D3L let P be a graded k-algebra. Then*

$$T\mathcal{D}ef_P \quad \text{and} \quad Inf(\mathcal{D}ef_P) = Der_k(P, P)$$

are finite dimensional if P is finitely generated over k.

Proof. We first deal with the infinitesimal automorphisms. Let $Q = P \otimes_k k[\epsilon]$. Then an element of $\mathrm{Inf}(\mathcal{D}ef_P)$ is given by an automorphism $\gamma = \mathrm{id} + \epsilon\delta : Q \to Q$ as above where now $\delta : P \to P$. The fact that γ is graded implies that δ is homogeneous of degree 0. The fact that γ is k-linear implies that δ is k-linear. The fact that γ is multiplicative implies that δ is a k-derivation. Conversely, given a k-derivation $\delta : P \to P$ homogeneous of degree 0, we obtain an automorphism $\gamma = \mathrm{id} + \epsilon\delta$ as above. Thus we see that

$$\mathrm{Inf}(\mathcal{D}ef_P) = \mathrm{Der}_k(P, P)$$

as predicted in the lemma. Clearly, if P is generated in degrees P_i, $0 \leq i \leq N$, then δ is determined by the linear maps $\delta_i : P_i \to P_i$ for $0 \leq i \leq N$ and we see that

$$\dim_k \mathrm{Der}_k(P, P) < \infty$$

as desired.

Date: stacks-project:examples-defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

To finish the proof of the lemma we show that there is a finite dimensional deformation space. To do this we choose a presentation

$$k[X_1, \ldots, X_n]/(F_1, \ldots, F_m) \longrightarrow P$$

of graded k-algebras where $\deg(X_i) = d_i$ and F_j is homogeneous of degree e_j. Let Q be any graded $k[\epsilon]$-algebra finite free in each degree which comes with an isomorphsm $\alpha : Q/\epsilon Q \to P$ so that (Q, α) defines an element of $T\mathcal{D}ef_P$. Choose a homogeneous element $q_i \in Q$ of degree d_i mapping to the image of X_i in P. Then we obtain

$$k[\epsilon][X_1, \ldots, X_n] \longrightarrow Q, \quad X_i \longmapsto q_i$$

and since $P = Q/\epsilon Q$ this map is surjective by Nakayama's lemma. A small diagram chase shows we can choose homogeneous elements $F_{\epsilon,j} \in k[\epsilon][X_1, \ldots, X_n]$ of degree e_j mapping to zero in Q and mapping to F_j in $k[X_1, \ldots, X_n]$. Then

$$k[\epsilon][X_1, \ldots, X_n]/(F_{\epsilon,1}, \ldots, F_{\epsilon,m}) \longrightarrow Q$$

is a presentation of Q by flatness of Q over $k[\epsilon]$. Write

$$F_{\epsilon,j} = F_j + \epsilon G_j$$

There is some ambiguity in the vector $(G_1, \ldots, G_m)$. First, using different choices of $F_{\epsilon,j}$ we can modify G_j by an arbitrary element of degree e_j in the kernel of $k[X_1, \ldots, X_n] \to P$. Hence, instead of $(G_1, \ldots, G_m)$, we remember the element

$$(g_1, \ldots, g_m) \in P_{e_1} \oplus \ldots \oplus P_{e_m}$$

where g_j is the image of G_j in P_{e_j}. Moreover, if we change our choice of q_i into $q_i + \epsilon p_i$ with p_i of degree d_i then a computation (omitted) shows that g_j changes into

$$g_j^{new} = g_j - \sum\nolimits_{i=1}^{n} p_i \partial F_j/\partial X_i$$

We conclude that the isomorphism class of Q is determined by the image of the vector $(G_1, \ldots, G_m)$ in the k-vector space

$$W = \mathrm{Coker}(P_{d_1} \oplus \ldots \oplus P_{d_n} \xrightarrow{(\frac{\partial F_j}{\partial X_i})} P_{e_1} \oplus \ldots \oplus P_{e_m})$$

In this way we see that we obtain an injection

$$T\mathcal{D}ef_P \longrightarrow W$$

Since W visibly has finite dimension, we conclude that the lemma is true. $\square$